Playlist Management — Technical Design Specification

Music application · PostgreSQL & REST API extension

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Abstract

This specification describes the backend extension that adds **playlist management** to the existing music catalog: new PostgreSQL tables, referential relationships to catalog **tracks**, and REST endpoints exposed by the Next.js application on Vercel. It defines the data model, entity relationships, API contract, deployment expectations, and a path toward authenticated users and role-based access. The design keeps catalog data normalized, avoids duplicate track metadata in playlists, and uses a junction table for many-to-many membership.

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1. Scope

In scope

- PostgreSQL schema additions: `playlists`, `playlist_tracks`.
- REST API under `/api` for creating and listing playlists, managing track membership, and administrative list/delete operations.
- Documentation of intended access rules (customer vs administrator) and interim behavior where full authentication is not yet wired.

Out of scope for this document's implementation phase

- End-user interface for playlists (handled in a separate front-end effort).
- Production user registration, session management, and middleware enforcement of roles (described as a forward-looking integration).

The existing `albums` and `tracks` tables remain the system of record for catalog content; this extension **adds** playlist structures that reference `tracks.id`.

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2. Capabilities and requirements

End users (customers)

- Create named playlists and associate catalog tracks with them.
- Add and remove tracks within playlists they control (once ownership is enforced server-side).
- List their own playlists when filtered by owner identity.

Administrators

- View all playlists (moderation and support).
- Delete any playlist when required for policy or data hygiene.

Implementation note

- "Songs" in the API and schema correspond to rows in `tracks` (integer primary keys, tied to `albums`).

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3. Data model

3.1 Source of truth

DDL for new objects:

playlists_schema.sql

Apply after `albums` and `tracks` exist. Runtime database: **PostgreSQL** (including managed Postgres on Vercel).

3.2 Table: `playlists`

Column	Type	Constraints
`id`	`UUID`	Primary key; default `gen_random_uuid()`
`name`	`VARCHAR(100)`	Not null
`owner_user_id`	`UUID`	Nullable; reserved for future FK to `users.id`
`created_at`	`TIMESTAMPTZ`	Not null; default `now()`

Index: `idx_playlists_owner_user_id` on `(owner_user_id)`.

3.3 Table: `playlist_tracks` (junction)

Column	Type	Constraints

Column	Type	Constraints
`playlist_id`	`UUID`	Not null; FK → `playlists(id)` **ON DELETE CASCADE**
`track_id`	`INTEGER`	Not null; FK → `tracks(id)` **ON DELETE CASCADE**
`added_at`	`TIMESTAMPTZ`	Not null; default `now()`

Primary key: `(playlist_id, track_id)` — unique membership per playlist/track pair.

Index: `idx_playlist_tracks_track_id` on `(track_id)`.

Extension: `CREATE EXTENSION IF NOT EXISTS pgcrypto;` (UUID generation).

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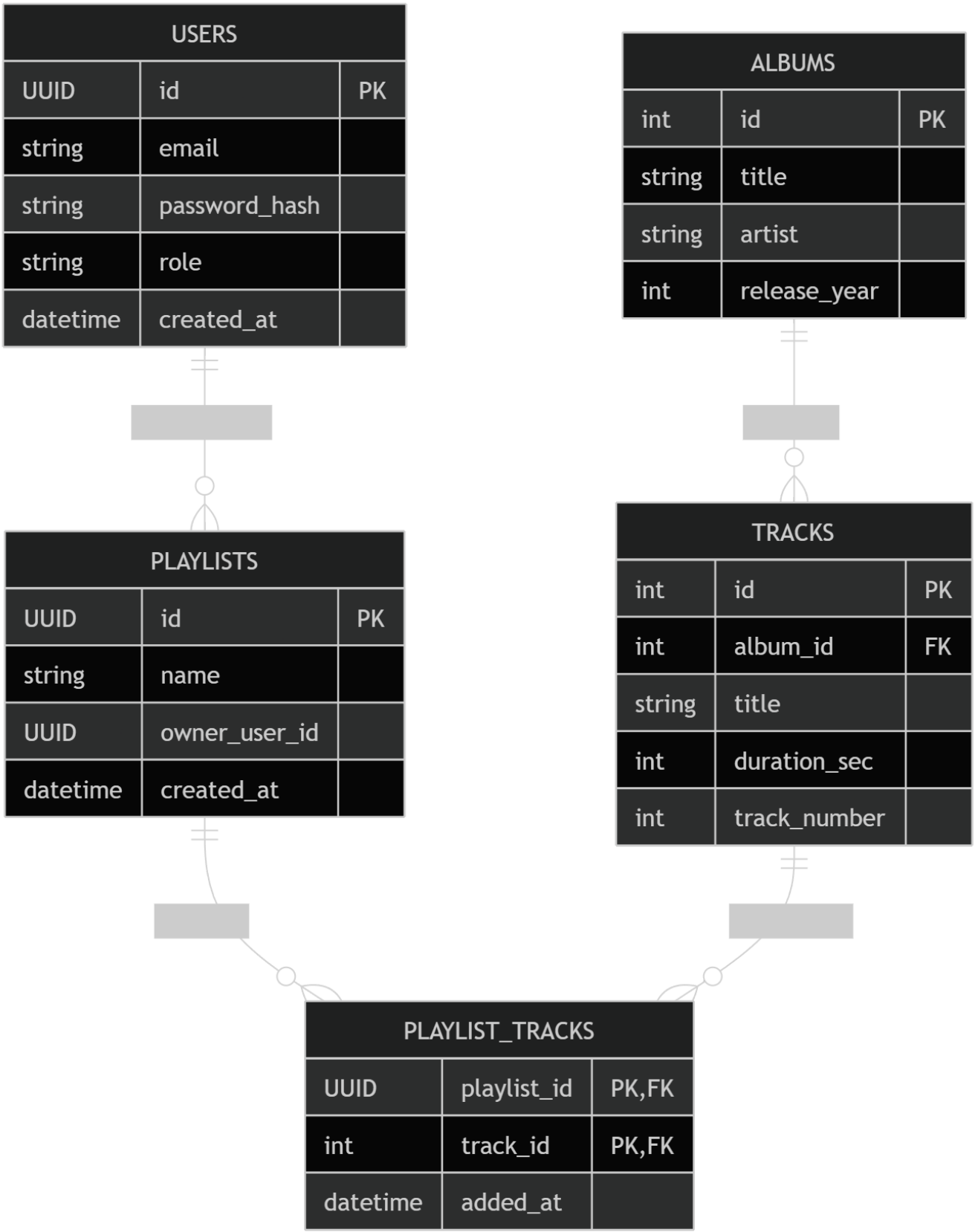
4. Entity-relationship documentation

Relationships

- **`albums`** → **`tracks`**: one-to-many (catalog).
- **`playlists`** ↔ **`playlist_tracks`** ↔ **`tracks`**: many-to-many; playlist membership references catalog tracks only by id.
- **`users`** (conceptual / future): one-to-many with **`playlists`** via `owner_user_id` once a `users` table and FK exist.

Diagram (crow's foot) — full model: future **`users`**, **`playlists`**, **`playlist_tracks`**, **`albums`**, **`tracks`**. UUIDs identify users and playlists; integers identify albums and tracks, consistent with the shipped SQL script.



Machine-readable summary (Mermaid) — same topology for version control and tooling:

```
erDiagram
  albums ||--o{ tracks : contains
  playlists ||--o{ playlist_tracks : has
  tracks ||--o{ playlist_tracks : in_playlist
  users }o--o{ playlists : owns_when_implemented
```

```
  albums {
    serial id PK
    varchar title
    varchar artist
    int year
  }
  tracks {
    serial id PK
    int album_id FK
    varchar title
    int number
  }
  playlists {
    uuid id PK
    varchar name
    uuid owner_user_id "nullable, FK to users later"
    timestamptz created_at
  }
  playlist_tracks {
    uuid playlist_id PK_FK
    int track_id PK_FK
    timestamptz added_at
  }
  users {
    uuid id PK "future"
  }
```

Related assets

- updated-er-diagram.png — primary diagram (embedded above).
- musicplayerUML.png, musicplayerER.png — earlier conceptual diagrams (archive).

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5. Design rationale

- **Normalization** — Track titles and album context remain in `tracks` / `albums`. The junction stores only ids and `added_at`, avoiding redundant catalog columns on each playlist row.
- **Many-to-many** — Standard junction pattern: a track may appear on many playlists; a playlist holds many tracks.
- **Uniqueness** — Composite primary key `(playlist_id, track_id)` prevents duplicate rows without application-only checks.
- **Referential integrity** — `ON DELETE CASCADE` on junction FKs removes membership when a playlist or catalog track is deleted, preventing orphaned links.
- **Performance** — Indexes support filtering playlists by owner and resolving “which playlists contain this track” when needed.

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6. Identity, ownership, and access control

Current state

- `owner_user_id` is nullable so the service can run before a **`users`** table exists. Playlists without an owner support integration testing; playlists with a UUID support owner-scoped behavior using interim client-supplied identifiers (e.g. query parameter or header) until sessions exist.

Target state

- Introduce **`users`** (e.g. `id UUID PRIMARY KEY`) and add a **foreign key** from `playlists.owner_user_id` to `users.id`.
- Derive the acting user from the authenticated session; do not rely on client-supplied owner ids alone for authorization.
- **Customer** operations: create and mutate only playlists owned by the signed-in user.
- **Administrator** operations: global playlist visibility and delete; enforce **`admin`** role in middleware or route guards.

Interim API behavior

- Where ownership is set on a playlist, mutating routes may require a matching `X-Owner-User-Id` header as a bridge until JWT/session integration is complete. Listing without an owner filter may return all playlists in non-production or test configurations; production should narrow by identity once auth is live.

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7. REST API

Conventions

- Base path: Next.js App Router, **`/api`**.
- JSON request and response bodies unless otherwise stated.
- Errors: typically `{ "error": "string" }`.

The **Authorization** column describes the **intended** production policy. Interim behavior may be more permissive during integration testing.

`POST /api/playlists`

Summary	Create a playlist.
Authorization	Authenticated end user.
Body	`{ "name": string }` — required, 1–100 characters. Optional: `"ownerUserId": string` (UUID) or `null`.
Success	`201 Created` — `id`, `name`, `ownerUserId`, `createdAt`, `trackCount` (e.g. `0`).
Errors	`400` validation; `500` server.

`GET /api/playlists`

Summary	List playlists; optionally restrict by owner.

Authorization	End user sees own data when filtered; unauthenticated access is not assumed for production.
Query	Optional `ownerUserId` (UUID). When set, returns playlists for that owner. When omitted, behavior may return all rows in controlled environments; lock down in production.
Success	`200 OK` — array of `{ id, name, ownerUserId, createdAt, trackCount }`.
Errors	`400` invalid UUID; `500` server.

`POST /api/playlists/{id}/tracks`

Summary	Add a catalog track to a playlist. `{id}` is playlist UUID.
Authorization	Playlist owner when `owner_user_id` is set.
Headers	`X-Owner-User-Id` must match owner when ownership applies (interim).
Body	`{ "trackId": number }` — `tracks.id`.
Success	`201 Created` — e.g. `{ playlistId, trackId }`.
Errors	`400`, `403`, `404`, `409` (duplicate membership), `500`.

`DELETE /api/playlists/{id}/tracks/{trackId}`

Summary	Remove a track from a playlist. `trackId` is `tracks.id`.
Authorization	Owner when ownership applies.
Headers	`X-Owner-User-Id` when required (interim).
Success	`204 No Content`.
Errors	`400`, `403`, `404`, `500`.

`GET /api/admin/playlists`

Summary	List all playlists (operations / moderation).
Authorization	Administrator.
Success	`200 OK` — same shape as unfiltered customer list.
Errors	`500`.

`DELETE /api/admin/playlists/{id}`

Summary	Delete a playlist by UUID; junction rows cascade.

Authorization	Administrator.
Success	`204 No Content`.
Errors	`400`, `404`, `500`.

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8. Implementation and operations

- Run **playlists_schema.sql** on each PostgreSQL environment (local and hosted) after base catalog tables exist.
- API route handlers live under `app/api/` in the Next.js repository; deploy with the Vercel project.
- Configure `POSTGRES_URL` or `DATABASE_URL` for the database where migrations were applied.
- Follow REST norms: plural resource collections, appropriate status codes (`201` create, `204` delete with empty body, `4xx` for client issues).

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9. Verification

Recommended validation before release:

- **Automated or manual API tests** — Exercise each endpoint (create playlist → add tracks → list → admin list → delete) against local and staged bases.
- **Postman collections** — Repository copies: `cst391-music-app/postman/CST391-Music-API-Local.postman_collection.json`, `cst391-music-app/postman/CST391-Music-API-Vercel.postman_collection.json`.
- **Recorded walkthrough** — Short demonstration of one handler's structure and successful calls against the deployed API supports onboarding and audit trails.

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10. References

- Grand Canyon University — CST-391 course materials and tutorials.
- Course milestone specification: CST-391 Milestone 4 — Database and API Implementation.
- Prior planning artifact: Project Proposal.md.
- NIST — Role-based access control (general background).

End of document.